

Project Design Phase
Proposed Solution Template

Date	18 June 2026
Team ID	
Project Name	ShopEZ
Maximum Marks	2 Marks

Proposed Solution Template:

Project team shall fill the following information in the proposed solution template.

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Small-to-medium retail vendors face massive technical debt, disconnected administration setups, high mobile site latency, and opaque invoicing checkouts.
2.	Idea / Solution description	A unified MERN application providing dynamic homepage visual configuration, fluid responsive styling, stateless JWT authentication, and a completely transparent checkout workflow.
3.	Novelty / Uniqueness	Dynamic Theme Injection: Injecting an .admin-theme class into the <body> to instantly re-map global CSS variables into a dark dashboard layout without template duplication. Flexible Database Proxying (makeModelProxy): A custom proxy interceptor pattern that recursively casts 24-character strings into native ObjectIds, allowing seamless swapping between in-memory mock testing and live MongoDB Atlas cloud instances.
4.	Social Impact / Customer Satisfaction	Democratizes local digital retail, allowing independent boutique merchants to shift physical inventories online within minutes and survive local foot-traffic shutdowns.
5.	Business Model (Revenue Model)	Self-Hosted SaaS: Monthly subscription model for custom subdomains, automated SSL management, and managed cloud clusters. Campaign / Banner Placements: Leasing promotional hero banners to third-party suppliers based on ad impressions. Premium Gateway Commissions: Collecting a 1.5% to 2.5% transaction cut on credit card gateway integrations.
6.	Scalability of the Solution	React Vite assets distributed via global CDNs; containerized stateless Express APIs deployed on AWS ECS/Google Cloud Run behind NGINX load balancers; MongoDB read-replication coupled with a Redis memory cache layer.